

Question 1

- (a) True
- (b) False
- (c) False
- (d) True

Question 2

- (a) First note that $*$ is clearly well-defined as $\varphi(g) \in G$. We first show that $\varphi_1 * (\varphi_2 * g) = (\varphi_1 \varphi_2) * g$ for all $\varphi_1, \varphi_2 \in \text{Aut}(G)$ and $g \in G$. So suppose $\varphi_1, \varphi_2 \in \text{Aut}(G)$ and $g \in G$. Now $\varphi_1 * (\varphi_2 * g) = \varphi_1 * \varphi_2(g) = \varphi_1(\varphi_2(g)) = (\varphi_1 \circ \varphi_2)(g) = (\varphi_1 \circ \varphi_2) * g = (\varphi_1 \varphi_2) * g$. Now we show that $1_{\text{Aut}(G)} * g = g$ for all $g \in G$. So let $g \in G$. But we know $1_{\text{Aut}(G)}$ is the identity map, id_G and $\text{id}_G * g = \text{id}_G(g) = g$ as required. So $*$ is a group action
- (b) $S(1_G) = \{\varphi \in \text{Aut}(G) : \varphi * 1_G = 1_G\}$. Now $\varphi * 1_G = \varphi(1_G)$, but since φ is an automorphism and thus a homomorphism, we get $\varphi(1_G) = 1_G$ for all $\varphi \in \text{Aut}(G)$ and so $S(1_G) = \text{Aut}(G)$

Question 3

By the Orbit-Stabiliser Theorem, $|G * x| = |G : S(x)|$. We know $S(x)$ is a subgroup of G , and so by Lagrange's Theorem, $|G : S(x)| = \frac{|G|}{|S(x)|}$ since G is finite. But then we get $|G * x| = \frac{|G|}{|S(x)|} \iff |G * x|(|S(x)|) = |G|$ and so since $|S(x)| \in \mathbb{Z}$, $|G * x|$ divides $|G|$